Counterfactual Stability Analysis of Classical Lane Perception

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Abstract

Autonomous perception pipelines are typically evaluated using per-frame metrics such as detection confidence and intersection-over-union, yet operate in a fundamentally temporal setting where successive outputs must be mutually consistent. We investigate whether realistic visual perturbations can induce hidden temporal instability that per-frame metrics miss. Using a classical lane perception pipeline on KITTI driving sequences, we apply synthetic cast shadows and horizontal motion blur, measuring five complementary stability metrics. Shadow perturbations preserve 100% *per-frame vanishing-point recovery while doubling frame-to-frame jitter (*$2.02\times$*) and inducing single-frame excursions of over* 200 *px—a failure mode invisible to static evaluation. Motion blur degrades recovery to* 67.5% *while disproportionately destabilizing angular geometry (*$6.43\times$ *baseline slope oscillation). Compounding analysis shows that instability manifests as transient events rather than continuous drift, and no single metric captures all failure modes. We argue that temporal inconsistency is itself a candidate uncertainty signal for autonomous perception.*

1. Introduction

Autonomous driving systems depend on visual perception pipelines that interpret road geometry frame by frame, such as detecting lane markings, estimating ego-trajectory, and tracking surrounding agents. These pipelines are commonly evaluated using static per-frame metrics such as intersection-over-union (IoU), recall, and detection confidence [4]. Yet autonomous systems do not operate frame by frame in isolation—they integrate perception outputs sequentially. Downstream planners and controllers are sensitive not just to whether a detection is correct, but to whether successive detections are *consistent*.

This distinction matters because per-frame and temporal failure modes can decouple. A perception pipeline may produce per-frame outputs that satisfy every static metric while exhibiting silent temporal inconsistency. For instance, when small frame-to-frame oscillations appear correct in isolation but, taken together, contradict any plausible physical trajectory. Such inconsistency is functionally a form of perceptual hallucination and is precisely the regime where overconfident downstream behavior is most likely.

We investigate this gap by introducing a counterfactual stability framework for evaluating classical lane perception under realistic environmental perturbations. Rather than studying adversarial attacks, which are well-characterized and often distributionally implausible, we apply semantically natural perturbations (cast shadows and motion blur) and ask whether they induce temporal instability *disproportionate to their effect on per-frame outputs*. Our central hypothesis is that traditional per-frame metrics systematically underestimate the impact of such perturbations, and that frame-to-frame stability metrics expose failure modes that static evaluation may not.

This work is centered around three key contributions. First, we present an end-to-end classical lane perception pipeline on the KITTI driving dataset [4], characterizing its baseline temporal jitter on unperturbed sequences. Second, we implement two physically motivated counterfactual perturbations and quantify their effect across five complementary stability metrics. Third, we analyze the *compounding behavior* of perturbation-induced instability over time, demonstrating that different perturbations produce qualitatively different failure signatures even when per-frame detection remains intact.

2. Related Work

2.1. Classical lane detection

Classical lane detection pipelines emerged in the 1990s and 2000s, combining low-level edge detection [2] with geometric primitives extracted via the Hough transform [3]. Aly [1] formalized a now-standard pipeline using inverse perspective mapping, Gaussian filtering, and RANSAC line fitting. While modern approaches have largely supplanted these methods for benchmark performance, classical pipelines remain widely used in safety-critical and resource-constrained settings due to their interpretability and minimal compute requirements. Our work builds on the canonical Canny–Hough–linear fit pipeline, treating it as an instrument for studying how visual perturbations propagate through interpretable geometric inference.

2.2. Modern lane detection

Since the late 2010s, deep learning has dominated lane detection benchmarks. SCNN [9] introduced spatial message-passing tailored to elongated lane structures, and subsequent attention-based methods such as LaneATT [10] pushed accuracy further on standard benchmarks. These approaches optimize per-frame detection metrics under the assumption that strong frame-level performance translates to robust system behavior. Here, we problematize this assumption: we show that even when per-frame outputs are preserved, temporal structure can degrade in ways invisible to static evaluation.

2.3. Robustness under realistic perturbations

A distinct literature studies robustness to natural rather than adversarial perturbations. Hendrycks and Dietterich [5] introduced a standardized benchmark of common corruptions (fog, motion blur, brightness changes) for image classification. Michaelis et al. [8] extended this framing to object detection on driving data, demonstrating that detection performance degrades substantially under physically plausible conditions. Our perturbations—synthetic cast shadows and directional motion blur—are drawn from this same family of semantically natural distortions, but we apply them in a temporally extended setting where the relevant outcome is not per-frame accuracy but trajectory-level consistency.

2.4. Temporal consistency and uncertainty

Several lines of work explicitly target the temporal dimension. Kendall and Gal [6] formalized aleatoric and epistemic uncertainty in deep vision systems, providing a framework that has been extended to driving perception. Temporal smoothing and Kalman-style filtering are routinely applied to lane outputs to suppress frame-to-frame jitter, but these methods often presuppose that jitter is noise rather than signal. Our contribution inverts that assumption; we treat temporal inconsistency as a candidate uncertainty signal in its own right and characterize how it responds to the aforementioned perturbations. To our knowledge, no prior work analyzes the *compounding behavior* of perturbation-induced instability in classical lane perception specifically, nor the differential sensitivity of multiple stability metrics to different perturbation types.

3. Methodology

3.1. Lane perception pipeline

We implement a classical lane perception pipeline operating on the left rectified color camera (`image_02`) of KITTI raw driving sequences. Each frame is converted to grayscale, smoothed with a 5×5 Gaussian, and processed by Canny edge detection (T_{lo}=50, T_{hi}=150). A fixed trapezoidal region of interest restricts subsequent processing to the road surface; ROI corners are hand-tuned per sequence to accommodate differences in camera framing.

Probabilistic Hough transform [7] extracts line segments from masked edges, retaining only segments with $|\text{slope}| > 0.4$ to exclude near-horizontal artifacts. Segments are partitioned by slope sign into left ($m < 0$) and right ($m > 0$) candidates, and a single line is fit to each group by weighted least squares with weights proportional to segment length. The vanishing point $\mathbf{v} = (v_x, v_y)$ is computed as the intersection of the two fitted lines.

The pipeline tracks only the two boundaries of the ego-occupied lane; multi-lane detection would require slope clustering and is outside our scope.

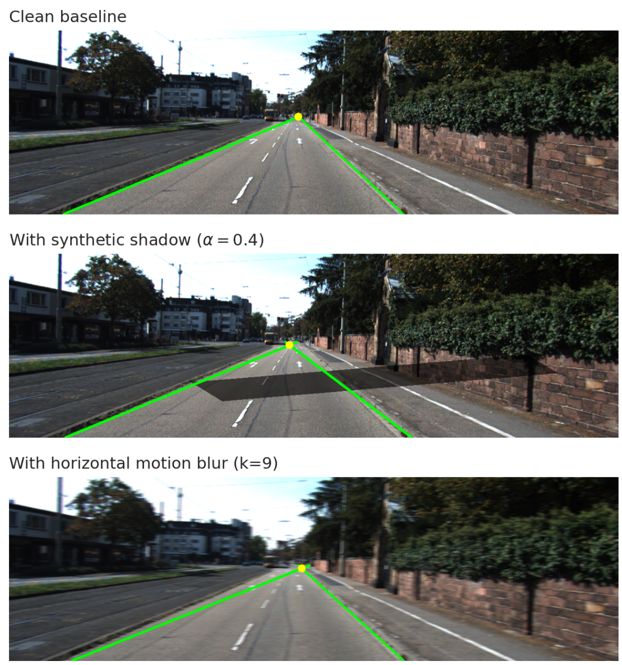

Figure 1. Pipeline output on a single representative frame under clean, shadow ($\alpha = 0.4$), and motion blur ($k = 9$) conditions. Detected lane boundaries shown in green, vanishing point in yellow. Both perturbations preserve a visually plausible lane geometry on this frame, yet (as we show in subsequent sections) destabilize temporal consistency in distinct ways.

3.2. Counterfactual perturbations

To probe complementary failure regimes (silent geometric drift versus partial detection collapse), we selected two different types of perturbations. Both are deterministic across the sequence, isolating the pipeline's response from input-side stochasticity (Figure 1).

Synthetic cast shadow. A fixed quadrilateral Π at the mid-distance road surface is multiplicatively darkened by $\alpha = 0.4$, simulating a cast shadow from an off-frame object. The polygon coordinates are scaled to each sequence's image dimensions but otherwise identical across frames within a sequence. We therefore expected per-frame lane detection to remain largely intact, since Canny edge gradients are still detectable in unshadowed regions. However, lane line positions should be biased where the shadow overlaps marking edges, producing temporal jitter without detection failure.

Horizontal motion blur. A 9×9 horizontal box-filter kernel is convolved with each frame, simulating directional smearing along the ego-motion axis. Kernel size and orientation are fixed across all frames. Horizontal motion blur attenuates the high-frequency content that Canny edge detection depends on. Here, we expected blur to degrade per-frame recovery directly, with secondary instability on frames where partial recovery occurred.

3.3. Stability metrics

We track the vanishing point as our primary stability signal because it integrates information from both lane boundaries into a single geometric quantity with direct physical interpretation as ego-heading direction. Compared to per-lane slope or lane-width estimates—which are sensitive to local fit noise on a single boundary—the vanishing point averages over both fits and is correspondingly more stable on clean sequences. We expect a small but non-zero VP jitter in the clean baseline arising from three sources: (i) discretization noise in Hough-segment endpoints, (ii) genuine ego-motion producing real frame-to-frame VP movement, and (iii) minor mismatches between the fixed ROI and the actual lane geometry as the vehicle moves within its lane.

This baseline jitter is the noise floor against which perturbation-induced instability must be measured; a perturbation effect is meaningful only when it elevates jitter substantially above this floor. Our measured perturbation effects fall in the 2–6$\times$ baseline range, supporting the conclusion that they reflect real pipeline degradation rather than stochastic variation around the noise floor.

The pipeline is evaluated across five metrics. Let $v_x[t]$ denote the vanishing point x-coordinate at frame t, and let $m_L[t], m_R[t]$ denote left and right lane slopes, respectively. We compute:

- **Recovery rate**: fraction of frames with a successfully recovered vanishing point
- **VP-x jitter**: $\mathbb{E}_t[|v_x[t{+}1] - v_x[t]|]$ over recovered consecutive frames
- **Slope oscillation**: $\frac{1}{2}(\mathbb{E}_t[|\Delta m_L|]+\mathbb{E}_t[|\Delta m_R|])$, capturing angular instability
- **2D VP drift**: $\mathbb{E}_t[\|\mathbf{v}[t{+}1] - \mathbf{v}[t]\|_2]$, the 2D analogue of VP-$x$ jitter
- **Heading deviation**: $\mathbb{E}_t[|v_x^{\text{pert}}[t] - v_x^{\text{clean}}[t]|]$, measuring trajectory-level bias from the clean baseline

Compounding behavior is analyzed via the cumulative drift curve $C[t] = \sum_{s\le t} |v_x[s{+}1] - v_x[s]|$ and the lagged autocorrelation of $|\Delta v_x|$.

3.4. Evaluation protocol

We evaluate on two independent KITTI raw sequences: `2011_09_26_drive_0002` (77 frames, primary) and `2011_09_29_drive_0026` (158 frames, validation), recorded on different calibration days. For each sequence, we run the pipeline under three conditions—clean, shadow, motion blur—yielding six pipeline traces in total. ROI geometry is the only parameter retuned between sequences; all other pipeline and perturbation parameters are held fixed.

4. Results

4.1. Baseline pipeline performance

On the primary sequence (drive_0002, 77 frames), the classical pipeline recovers a vanishing point on every frame (100% recovery; Figure 2). Mean frame-to-frame VP-x jitter is 9.55 px and the standard deviation of v_x is 24.99 px. The recovered VP track exhibits a smooth low-frequency drift consistent with ego-motion through the scene, confirming that the pipeline produces a coherent temporal signal rather than per-frame noise, which is a necessary precondition for any temporal-stability analysis.

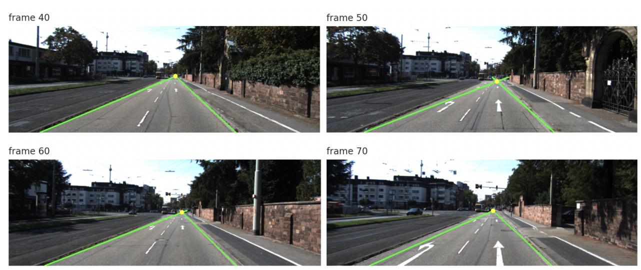

Figure 2. Pipeline output on four consecutive frames of the clean drive_0002 sequence (frames 40, 50, 60, 70). The vanishing point traces a smooth low-frequency trajectory consistent with ego-motion through the scene.

4.2. Perturbation effects: shadow and motion blur

Applying the synthetic shadow perturbation, per-frame recovery remains at 77/77: any per-frame detection-confidence metric would declare the pipeline unaffected. Yet mean VP-x jitter rises to 19.30 px (a 2.02$\times$ increase) and the standard deviation to 30.03 px (Figure 3). The shadow track tracks the clean baseline closely for most of the sequence but exhibits a single-frame VP-x excursion of $>$ 200 px at frame 57—a temporal failure that is plainly anomalous in context yet invisible to any static frame-level evaluation. We return to this example in Section 4.5.

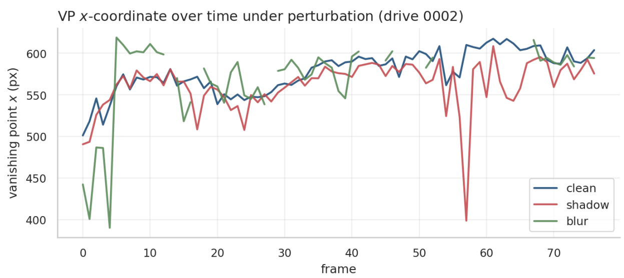

Figure 3. Vanishing point x-coordinate over time on drive_0002 under clean, shadow, and motion blur conditions. Per-frame VP recovery is 100% for clean and shadow; blur drops to 67.5% (gaps in the green trace). The shadow trace exhibits a > 200 px excursion at frame 57.

Motion blur produces a categorically different failure mode. Per-frame recovery drops to $52/77$ (67.5%): the pipeline outright fails to recover a vanishing point on a third of frames. Among recovered frames, mean VP-x jitter rises to 22.95 px ($2.40\times$ clean). The two perturbations therefore occupy distinct failure regimes: shadow induces silent temporal instability without per-frame detection failure, while motion blur induces both partial detection collapse and elevated instability on the frames where detection survives.

4.3. Metrics expose different failure signatures

Examining the full set of stability metrics (Table 1) reveals that no single metric captures both perturbations equally well. Slope oscillation, which measures angular instability of the individual lane line fits, increases only $1.80\times$ under shadow but $6.43\times$ under motion blur. In contrast, VP-x jitter is similarly elevated under both perturbations ($2.02\times$ and $2.40\times$, respectively). Motion blur therefore degrades the angular geometry of lane fits much more severely than the position of their intersection point, a finding that VP-based metrics alone would obscure. This suggests that uncertainty signals for autonomous perception should integrate multiple geometric dimensions: relying on VP-based monitoring alone would systematically underestimate blur's impact, while relying on slope-based monitoring alone would underestimate shadow's.

Metric	Clean	Shadow	Blur
Recovery rate	77/77	77/77	52/77
VP-x jitter (px)	9.55	19.30	22.95
Slope oscillation	0.030	0.054	0.193
2D VP drift (px)	11.61	22.22	27.97
Heading deviation (px)	0.00	18.50	23.93

Table 1. Stability metrics on drive_0002. Per-frame detection is preserved under shadow yet temporal stability degrades; motion blur degrades both.

4.4. Compounding behavior

We further examine whether perturbation-induced instability *accumulates* over a sequence—a question raised by the temporal-extension premise of our hypothesis. The cumulative drift curve (Figure 4) shows that neither perturbation exhibits monotonic super-linear compounding. Shadow tracks the clean baseline closely until a discrete instability event at frame 57 contributes ~ 500 px to its cumulative total in three frames; thereafter it resumes a roughly clean-baseline slope. Motion blur shows a sharp early ramp (~ 450 px accumulated in the first five frames) followed by a slope comparable to clean. Neither perturbation produces continuous error accumulation.

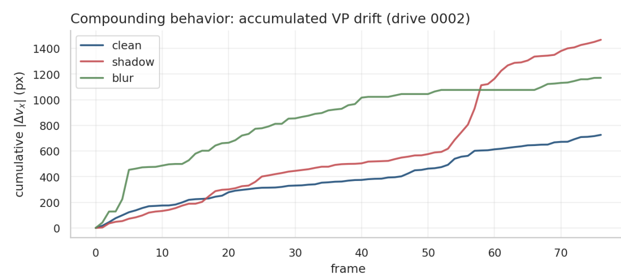

Figure 4. Cumulative absolute VP drift over the sequence. Neither perturbation produces monotonic super-linear compounding.

Lagged autocorrelation analysis (Figure 5) corroborates this picture. All three conditions show positive autocorrelation at lag 1 (0.36 clean, 0.55 shadow, 0.21 blur), indicating short-range temporal clustering of unstable frames, but signals decay to near zero by lag 6 for all conditions. Shadow exhibits the most temporally structured instability (0.36 mean autocorrelation over lags 1–4), while blur is the least structured. The implication is that perturbation effects manifest as transient instability events, bursts, rather than as sustained drift, and effective uncertainty signals should be sensitive to such events rather than smoothing over them.

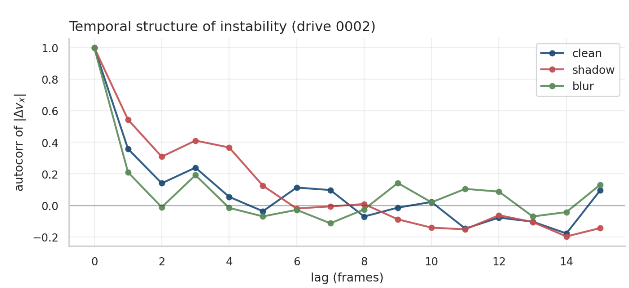

Figure 5. Lagged autocorrelation of $|\Delta v_x|$. All conditions show positive autocorrelation at lag 1, decaying to near zero by lag 6. Shadow exhibits the most temporally structured instability; blur is closest to white noise. Instability is locally clustered but does not exhibit long-range memory.

4.5. Per-frame undetectable failures: frame 57

Figure 6 makes the per-frame–vs.–temporal decoupling concrete. Frames 56, 57, and 58 of the shadow-perturbed sequence each produce a successfully recovered vanishing point and a per-frame detection that would satisfy standard accuracy metrics. Yet the frame-57 detection is plainly wrong on inspection: the fitted lane lines diverge into geometrically implausible configurations and the vanishing point is pulled $\sim$ 200 px below its surrounding-frame trajectory. The pipeline's per-frame outputs offer no signal that frame 57 is anomalous; the temporal context exposes it.

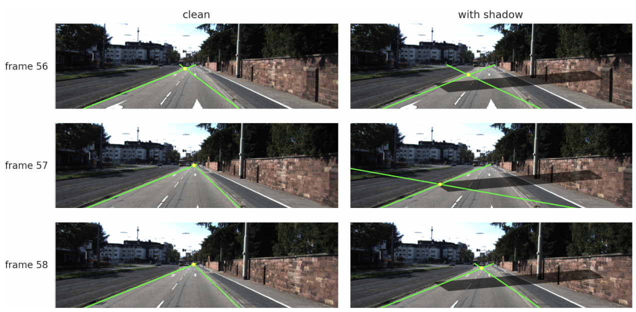

Figure 6. Frames 56, 57, 58 of drive_0002 under clean (left) and shadow (right) conditions.

4.6. Cross-sequence validation

To assess whether these findings generalize beyond a single drive, we evaluated the pipeline on a second independent sequence (drive_0026, 158 frames, recorded on a different calibration day). Perturbation effects qualitatively replicate: both shadow and blur increase every stability metric over their respective clean baselines on the new sequence (Figure 7). However, the magnitudes of perturbation effects vary substantially between sequences—and so does the relative ranking of metric sensitivities. Shadow's effect on VP-x jitter is $2.02\times$ on drive_0002 but $4.53\times$ on drive_0026; blur's recovery rate is 67.5% on drive_0002 but 99.4% on drive_0026, reflecting scene-dependent vulnerability to specific perturbations. This is a more nuanced finding than naive replication would have produced: perturbation-induced instability is a real phenomenon but its magnitude and metric profile depend on scene content. This motivates the use of multiple complementary metrics rather than any single uncertainty signal.

5. Conclusion

We presented a counterfactual stability framework for evaluating classical lane perception under realistic visual perturbations. By applying physically plausible cast shadows and motion blur to KITTI driving sequences and measuring five complementary stability metrics, we demonstrated that per-frame detection metrics systematically miss a class of failures that are clearly visible in the temporal dimension. Synthetic shadows induced a $2.02\times$ increase in frame-to-frame vanishing-point jitter and a $>$ 200 px single-frame excursion at frame 57 of drive_0002, yet preserved 100% per-frame detection recovery. Motion blur produced a categorically different signature, degrading recovery to 67.5% while disproportionately destabilizing angular geometry ($6.43\times$ baseline slope oscillation). Cross-sequence validation confirmed that these effects are real and directional, though their magnitudes are scene-dependent.

Two implications follow. First, temporal inconsistency is not noise to be smoothed away but a candidate uncertainty signal in its own right; it carries information about perturbation impact that per-frame metrics do not. Second, no single stability metric captures every perturbation: effective monitoring requires complementary measurements across position-space, angle-space, and trajectory-space. Compounding behavior analysis further suggests that uncertainty signals should be sensitive to transient instability *events* rather than to running averages, which would smooth over the single-frame anomalies most diagnostic of failure.

Several limitations bound these conclusions, including: the pipeline tracks only two boundaries of the ego-occupied lane, ROI geometry is hand-tuned per sequence, and we evaluated on two sequences from a single dataset. Natural extensions include applying this to learned perception pipelines, characterizing additional perturbation types (lane fading, partial occlusion, low-light), and testing whether temporal-stability monitoring can be operationalized as a real-time confidence signal for downstream planning. The decoupling between per-frame correctness and temporal consistency, we believe, deserves more attention than current evaluation practice affords it.

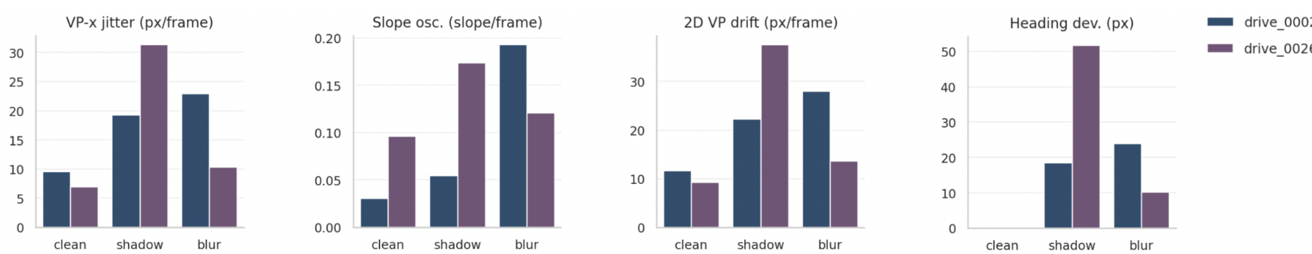

Figure 7. Stability metrics across both sequences and all three conditions. Perturbation effects are qualitatively present in both sequences.

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